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-- Experiment 2 & 3: SQL Constraints & Relational Model
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-- Create the Employee table
CREATE TABLE Employee (
    Fname VARCHAR(20) NOT NULL,
    Minit CHAR(1),
    Lname VARCHAR(30) NOT NULL,
    Ssn CHAR(9) PRIMARY KEY,
    Bdate DATE,
    Address VARCHAR(100),
    Gender CHAR(1),
    Salary DECIMAL(10,2),
    SUPERVISOR_SSN CHAR(9),
    Dno INT NOT NULL
);

-- Insert data into the Employee table
INSERT INTO Employee (Fname, Minit, Lname, Ssn, Bdate, Address, Gender, Salary,
SUPERVISOR_SSN, Dno) VALUES ('John', 'B', 'Smith', '123456789', TO_DATE
('1965-01-09', 'YYYY-MM-DD'), '731 Fondren, Houston, TX', 'M', 30000.00, '
333445555', 5);
INSERT INTO Employee (Fname, Minit, Lname, Ssn, Bdate, Address, Gender, Salary,
SUPERVISOR_SSN, Dno) VALUES ('Franklin', 'T', 'Wong', '333445555', TO_DATE
('1955-12-08', 'YYYY-MM-DD'), '638 Voss, Houston, TX', 'M', 40000.00,
'888665555', 5);
INSERT INTO Employee (Fname, Minit, Lname, Ssn, Bdate, Address, Gender, Salary,
SUPERVISOR_SSN, Dno) VALUES ('Alicia', 'J', 'Zelaya', '999887777', TO_DATE
('1968-01-19', 'YYYY-MM-DD'), '3321 Castle Spring, TX', 'F', 25000.00, '
987654321', 4);
INSERT INTO Employee (Fname, Minit, Lname, Ssn, Bdate, Address, Gender, Salary,
SUPERVISOR_SSN, Dno) VALUES ('Jennifer', 'S', 'Wallace', '987654321', TO_DATE
('1941-09-20', 'YYYY-MM-DD'), '291 Berry Bellaire, TX', 'F', 43000.00,
'888665555', 4);
INSERT INTO Employee (Fname, Minit, Lname, Ssn, Bdate, Address, Gender, Salary,
SUPERVISOR_SSN, Dno) VALUES ('Ramesh', 'K', 'Narayan', '666884444', TO_DATE
('1962-09-15', 'YYYY-MM-DD'), '975 Pine Oak, Humble, TX', 'M', 38000.00,
'333445555', 5);
INSERT INTO Employee (Fname, Minit, Lname, Ssn, Bdate, Address, Gender, Salary,
SUPERVISOR_SSN, Dno) VALUES ('Joyce', 'A', 'English', '453453453', TO_DATE
('1972-07-31', 'YYYY-MM-DD'), '5631 Rice, Houston, TX', 'F', 25000.00,
'333445555', 6);
INSERT INTO Employee (Fname, Minit, Lname, Ssn, Bdate, Address, Gender, Salary,
SUPERVISOR_SSN, Dno) VALUES ('Ahmad', 'V', 'Jabbar', '987987987', TO_DATE
('1969-03-29', 'YYYY-MM-DD'), '980 Dallas, Houston, TX', 'M', 25000.00, '
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987654321', 4);
INSERT INTO Employee (Fname, Minit, Lname, Ssn, Bdate, Address, Gender, Salary,
SUPERVISOR_SSN, Dno) VALUES ('James', 'E', 'Borg', '888665555', TO_DATE
('1937-11-10', 'YYYY-MM-DD'), '450 Stone, Houston, TX', 'M', 55000.00, NULL, 1)
SELECT Fname, Minit, Lname, Ssn, Bdate, Address, Gender, Salary, SUPERVISOR_SSN
Dno FROM Employee;

-- Create the Department table
CREATE TABLE Department (
    Dname VARCHAR(15) NOT NULL,
    Dnumber INT PRIMARY KEY,
    Mgr_ssn CHAR(9) NOT NULL,
    Mgr_start_date DATE,
    FOREIGN KEY (Mgr_ssn) REFERENCES Employee(Ssn)
);
DESC Department;
-- Insert data into Department
INSERT INTO Department (Dname, Dnumber, Mgr_ssn, Mgr_start_date) VALUES
('Research', 5, '333445555', TO_DATE('1988-05-22', 'YYYY-MM-DD'));
INSERT INTO Department (Dname, Dnumber, Mgr_ssn, Mgr_start_date) VALUES
('Administration', 4, '987654321', TO_DATE('1995-01-01', 'YYYY-MM-DD'));
INSERT INTO Department (Dname, Dnumber, Mgr_ssn, Mgr_start_date) VALUES
('Headquarters', 1, '888665555', TO_DATE('1981-06-19', 'YYYY-MM-DD'));
SELECT Dname, Dnumber, Mgr_ssn, Mgr_start_date FROM Department;

-- Create the Dept_locations table
CREATE TABLE Dept_locations (
    Dnumber INT NOT NULL,
    Dlocation VARCHAR(15) NOT NULL,
    FOREIGN KEY (Dnumber) REFERENCES Department(Dnumber)
);
DESC Dept_locations;
-- Insert data into Dept_locations
INSERT INTO Dept_locations (Dnumber, Dlocation) VALUES (5, 'Houston');
INSERT INTO Dept_locations (Dnumber, Dlocation) VALUES (4, 'Stafford');
INSERT INTO Dept_locations (Dnumber, Dlocation) VALUES (5, 'Bellaire');
INSERT INTO Dept_locations (Dnumber, Dlocation) VALUES (1, 'Sugarland');
INSERT INTO Dept_locations (Dnumber, Dlocation) VALUES (5, 'Houston');
SELECT Dnumber, Dlocation FROM Dept_locations;

-- Create the Project table
CREATE TABLE Project (
    Pname VARCHAR(15) NOT NULL UNIQUE,
    Pnumber INT PRIMARY KEY NOT NULL,
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    Plocation VARCHAR(15),
    Dnum INT NOT NULL,
    FOREIGN KEY (Dnum) REFERENCES Department(Dnumber)
);
DESC Project;
-- Insert data into Project
INSERT INTO Project (Pname, Pnumber, Plocation, Dnum) VALUES
('ProductX', 1, 'Bellaire', 5);
INSERT INTO Project (Pname, Pnumber, Plocation, Dnum) VALUES
('ProductY', 2, 'Sugarland', 5);
INSERT INTO Project (Pname, Pnumber, Plocation, Dnum) VALUES
('ProductZ', 3, 'Houston', 5);
INSERT INTO Project (Pname, Pnumber, Plocation, Dnum) VALUES
('Computerization', 10, 'Stafford', 4);
INSERT INTO Project (Pname, Pnumber, Plocation, Dnum) VALUES
('Reorganization', 20, 'Houston', 1);
INSERT INTO Project (Pname, Pnumber, Plocation, Dnum) VALUES
('Newbenefits', 30, 'Stafford', 4);
SELECT Pname, Pnumber, Plocation, Dnum FROM Project;

-- Create the Works_on table
CREATE TABLE Works_on (
    Essn CHAR(9) NOT NULL,
    Pno INT NOT NULL,
    Hours DECIMAL(3,1) NOT NULL,
    PRIMARY KEY (Essn, Pno),
    FOREIGN KEY (Essn) REFERENCES Employee(Ssn),
    FOREIGN KEY (Pno) REFERENCES Project(Pnumber)
);
DESC Works_on;
-- Insert data into Works_on
INSERT INTO Works_on (Essn, Pno, Hours) VALUES ('123456789', 1, 32.5);
INSERT INTO Works_on (Essn, Pno, Hours) VALUES ('123456789', 2, 7.5);
INSERT INTO Works_on (Essn, Pno, Hours) VALUES ('666884444', 3, 40.0);
INSERT INTO Works_on (Essn, Pno, Hours) VALUES ('453453453', 1, 20.0);
INSERT INTO Works_on (Essn, Pno, Hours) VALUES ('453453453', 2, 20.0);
INSERT INTO Works_on (Essn, Pno, Hours) VALUES ('333445555', 2, 10.0);
INSERT INTO Works_on (Essn, Pno, Hours) VALUES ('333445555', 3, 10.0);
INSERT INTO Works_on (Essn, Pno, Hours) VALUES ('333445555', 10, 10.0);
INSERT INTO Works_on (Essn, Pno, Hours) VALUES ('333445555', 20, 10.0);
INSERT INTO Works_on (Essn, Pno, Hours) VALUES ('999887777', 30, 30.0);
INSERT INTO Works_on (Essn, Pno, Hours) VALUES ('999887777', 10, 10.0);
INSERT INTO Works_on (Essn, Pno, Hours) VALUES ('987987987', 10, 35.0);
INSERT INTO Works_on (Essn, Pno, Hours) VALUES ('987987987', 30, 5.0);
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INSERT INTO Works_on (Essn, Pno, Hours) VALUES ('987654321', 30, 20.0);
INSERT INTO Works_on (Essn, Pno, Hours) VALUES ('987654321', 20, 15.0);
INSERT INTO Works_on (Essn, Pno, Hours) VALUES ('888665555', 20, 0.0);
SELECT Essn, Pno, Hours FROM Works_on;

-- Create the Dependent table
CREATE TABLE Dependent(
    Essn CHAR(9) NOT NULL,
    Dependent_name VARCHAR(30) NOT NULL,
    Gender CHAR(1),
    Bdate DATE NOT NULL,
    Relationship VARCHAR(15),
    PRIMARY KEY (Essn, Dependent_name),
    FOREIGN KEY (Essn) REFERENCES Employee(Ssn)
);
DESC Dependent;
-- Insert data into Dependent
INSERT INTO Dependent(Essn, Dependent_name, Gender, Bdate, Relationship) VALUES
('333445555', 'Alice', 'F', TO_DATE('1986-04-05', 'YYYY-MM-DD'), 'Daughter');
INSERT INTO Dependent(Essn, Dependent_name, Gender, Bdate, Relationship) VALUES
('333445555', 'Theodore', 'M', TO_DATE('1983-10-25', 'YYYY-MM-DD'), 'Son');
INSERT INTO Dependent(Essn, Dependent_name, Gender, Bdate, Relationship) VALUES
('333445555', 'Joy', 'F', TO_DATE('1958-05-03', 'YYYY-MM-DD'), 'Spouse');
INSERT INTO Dependent(Essn, Dependent_name, Gender, Bdate, Relationship) VALUES
('987654321', 'Abner', 'M', TO_DATE('1942-02-28', 'YYYY-MM-DD'), 'Spouse');
INSERT INTO Dependent(Essn, Dependent_name, Gender, Bdate, Relationship) VALUES
('123456789', 'Michael', 'M', TO_DATE('1988-01-01', 'YYYY-MM-DD'), 'Son');
INSERT INTO Dependent(Essn, Dependent_name, Gender, Bdate, Relationship) VALUES
('123456789', 'Alice', 'F', TO_DATE('1988-12-30', 'YYYY-MM-DD'), 'Daughter');
INSERT INTO Dependent(Essn, Dependent_name, Gender, Bdate, Relationship) VALUES
('123456789', 'Elizabeth', 'F', TO_DATE('1967-05-05', 'YYYY-MM-DD'), 'Spouse');
SELECT Essn, Dependent_name, Gender, Bdate, Relationship FROM Dependent;

-- Commit the changes for future experiments
COMMIT;
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